

# 2026-06-10-llms-access-master-roadmap

## Master Roadmap — Models-Access Refinement & Agents-Work Enforcement

| Field          | Value                                                                                                                                                                |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Revision       | 3                                                                                                                                                                    |
| Last modified  | 2026-06-10                                                                                                                                                           |
| Status summary | PLANNING COMPLETE — 6 analyses + 5 SP plans written; constitution synced; SP3 plan + bundle code-review in flight; awaiting operator on decision register (G-1..G-5) |
| Source request | docs/requests/<br>helix_code_request_llms_access.txt                                                                                                                 |
| Delivery model | <b>Independent sub-programs</b> (operator decision 2026-06-10) — this doc is the index/roadmap; each sub-program gets its own spec → plan → release cycle            |
| Authority      | Cascades from constitution/ + root CLAUDE.md/CONSTITUTION.md; anti-bluff §11.4 family governs every closure                                                          |

**Anti-bluff note (§11.4 / §11.4.123):** This roadmap is a *plan*, not a claim of done work. No phase below is “complete” until it carries captured runtime evidence per §11.4.5 / §11.4.69 / §11.4.107. The brainstorming HARD-GATE applies: **no implementation, fork creation, or push happens until each sub-program’s design+plan is approved.**

---

### 0. Operator decisions baked in (2026-06-10)

- Fork strategy:** Fork ALL ~50 cli\_agents/ \* third-party submodules **now** under the vasic-digital org with prefix caf- (cli-agent-fork) → e.g. git@github.com:vasic-digital/caf-claude-code.git. Swap each submodule’s URL to our fork. Auto-merge-from-upstream for every fork.
  - Delivery:** Independent sub-programs; this is the master index.
  - Bridge exposure: One provider per CLI agent** — each working CLI agent surfaces its own dynamically-discovered model list + power-features (Vision / Generative / RAG / Memory / MCP / LSP / ACP).
  - New “ways-of-working” submodules** (parallel-agents, dynamic-flows, ...): under the HelixDevelopment org.
  - Constitution DESCOPED (2026-06-10):** no governance-rule authoring, no constitution push — **sync only** (fetch+pull to current). Reusable fork/merge scripts live **local to HelixCode** (scripts/caf/), not in constitution/scripts/.
-

# 1. Architecture grounding (verified from repo, 2026-06-10)

- **HelixAgent is the “root”** the request extends — `submodules/helix_agent/internal/` already contains `ensemble, llm, router, verifier, providers, modelsdev, models, cli, mcp, rag, memory, agentic, agents, concurrency, planning, background`. (Verified by ls.)
- **LLMsVerifier** (`submodules/llms_verifier`) is the single source of truth for working models (CONST-036/040) — all dynamic discovery flows through it; hardcoded model lists are violations.
- **Inner Go app** (`helix_code/`) has `internal/llm/*_provider.go` (`anthropic, azure, bedrock, copilot, deepseek, gemini, groq, ...`), `internal/provider`, `internal/providers`, `internal/agent`.
- **Constitution scripts** live in `constitution/scripts/` (`codegraph, action-prefix, subagent-tiering, token-accounting, workable-items, hooks`). **No fork-maintenance scripts exist yet** — Sub-program SP0 adds them.
- **~50 CLI agents** under `cli_agents/` as third-party submodules (verified `.gitmodules`).

Per-workstream evidence-backed analyses (produced by 6 parallel subagents, 2026-06-10) land in `docs/superpowers/specs/analysis/2026-06-10-{A..F}-*.md` and feed the detailed sub-program specs.

---

## 1a. Real defects surfaced during planning (evidence-backed — to be fixed inside the relevant SP, RED-first §11.4.115)

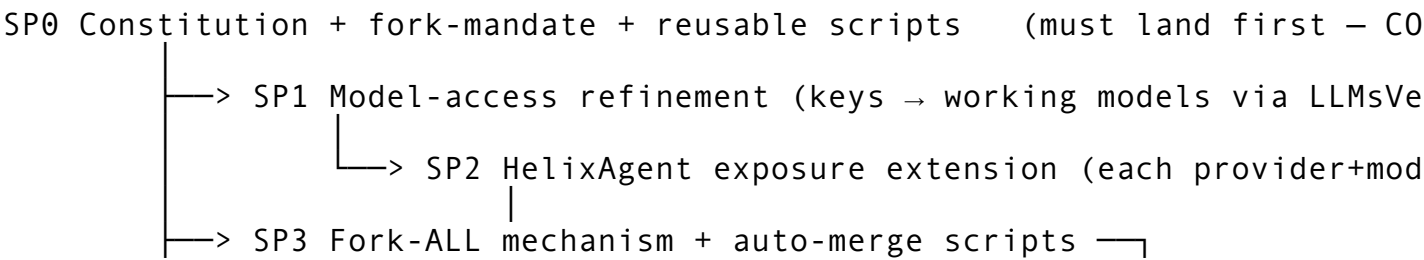
These are concrete §11.4 anti-bluff / CONST violations found by the analysis subagents with `file:line` evidence — independent of the new features, they are pre-existing defects:

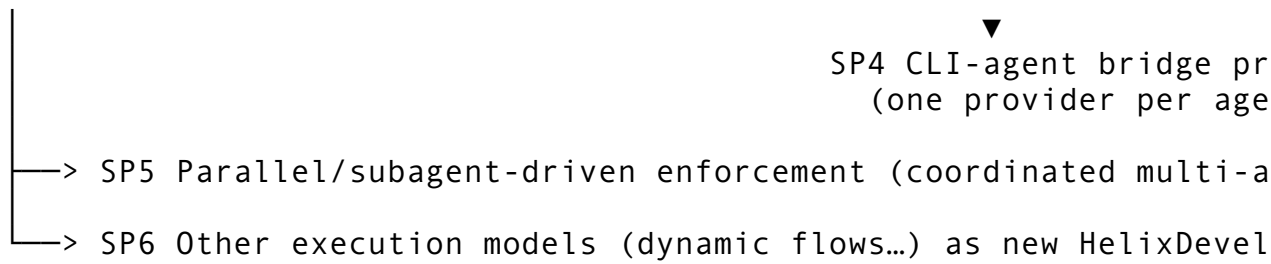
| ID  | Defect                                                                                               | Evidence                                                            | Owner SP                         |
|-----|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------|
| D-1 | <code>/v1/completion/</code> models returns a <b>hardcoded 3-model list</b>                          | <code>helix_agent/internal/handlers/completion.go:406</code>        | SP2<br>(CONST-036/<br>BLUFF-002) |
| D-2 | CLI <code>handleListModel</code> s lists failed/pending models as <b>available</b>                   | <code>helix_code/.../cmd/cli/main.go:1361</code>                    | SP1                              |
| D-3 | <code>secrets.LoadAPIKeys</code> is <b>DEAD code</b> (never called in prod)                          | <code>helix_code/internal/secrets/loader.go:30</code>               | SP1                              |
| D-4 | “Working-model” filter (Verified $\wedge$ <code>score ≥ min</code> ) <b>loaded but never applied</b> | <code>adapter.go:175</code>                                         | SP1                              |
| D-5 | <b>Hardcoded provider model lists</b> (OpenAI/Anthropic/DeepSeek/Mistral +~8)                        | <code>openai_provider.go:203, anthropic_provider.go:205, ...</code> | SP1<br>(CONST-036)               |
| D-6 |                                                                                                      |                                                                     |                                  |

| ID   | Defect                                                                                          | Evidence                                                                                              | Owner SP                                              |
|------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
|      | Stub CLI-agent packages return <b>hardcoded strings</b> , <b>never exec</b> the binary          | helix_agent/internal/clis/agents/qwencode/qwencode.go:101/115/137                                     | SP4 (CONST-035)                                       |
| D-7  | helix_code/go.mod lacks replace dev.helix.agent → can't import the real clis/agent ic substrate | helix_code/go.mod                                                                                     | SP4/SP5                                               |
| D-8  | Constitution CLAUDE.md Revision-table drift                                                     | constitution/CLAUDE.md                                                                                | SP0 resolved by sync (now Rev 21 @ f26368b)           |
| D-9  | <b>Pooled dispatch methods are stubs too</b> — return "<Agent> execution completed", no exec    | helix_agent/internal/clis/instance_manager.go:906-942 (executeAider/ClaudeCode/Codex/Cline/OpenHands) | ✅ FIXED (SP4, real os/exec, pin guards GREEN)         |
| D-6  | qwencode stub returns templated strings, never execs                                            | helix_agent/.../qwencode/qwencode.go:101                                                              | ✅ FIXED (SP4, real os/exec)                           |
| D-10 | 2 pre-existing flaky tests (nanosecond ID-collision)                                            | helix_agent TestCompletionHandler_IDGeneration, TestGenerateTeamID                                    | open (pre-existing, low-pri)                          |
| D-11 | pre-existing "For now, allow all types" bluff                                                   | helix_agent/internal/clis/instance_manager.go:389 (IsAgentTypeAvailable)                              | open (SP4 follow-up)                                  |
| D-12 | ~34 more execute* dispatch methods still stubbed                                                | helix_agent/internal/clis/instance_manager.go (executeKiro... executeLlamaCode)                       | open (SP4 follow-up wave)                             |
| D-13 | CONST-066 doc-export gap: 95+ docs/**/*.md (+ new specs/research) lack .html/.pdf siblings      | repo-wide docs/**                                                                                     | open (D-13; F1 hook scoped too leniently — reconcile) |

## 2. Sub-program decomposition

Each sub-program (SP) is independent: own spec, own plan, own evidence-backed release. Dependency order shown.





SP7 Testing & QA — cross-cutting, binds every SP (100% test-type coverage)

## SP0 — Constitution SYNC ONLY (DESCOPED per operator 2026-06-10)

**Operator decision 2026-06-10:** “We dont have need to change the constitution Submodule at all... just make constitution Submodule up to date, fetch and pull.” → **No §11.4.N authoring, no constitution push.** The reusable fork/merge scripts relocate to `scripts/caf/` **local to HelixCode** (not `constitution/scripts/`). Analysis-A’s drafted anchor is archived, unused (and §11.4.142 was in any case already taken upstream by the Universal-code-review mandate).

**Status: DONE (this session).** - Fetched all 8 constitution remotes; **fast-forwarded** 60e2d66 → f26368b **on main** (0 ahead / 4 behind / ancestor → no loss, §11.4.113 no-force). -   
Recursive check: **no nested constitution submodules** (root-only, CONST-051 clean);  
submodules/\*/CONSTITUTION.md are cascade file-copies. - New anchors now in  
force: **§11.4.142 Universal code-review** (every change → independent review before commit/  
build — applies to all SP implementation), **§11.4.143 video-journey** (N/A to HelixCode). -   
**Pending follow-up:** bump superproject submodule pointer 60e2d66 → f26368b (a meta-  
repo commit; itself subject to §11.4.142 review) — batch with next commit. - **Requested but  
not yet run** (request “Before we begin”): CONST-055 post-pull full-rule validation sweep across  
project + submodules; correct any violation found. → tracked as **SP0.b**.

### Original SP0 (superseded) — Constitution mandate + scripts

**Goal:** Land the new mandatory rule + generic reusable scripts in the constitution submodule BEFORE building on them (CONST-049 ordering). Feeds: analysis ... - A - .... (*Superseded by the descope above; the script designs from analysis-A/E are reused but land in scripts/caf/ local to HelixCode under SP3.*)

- **P0.1 Fetch/pull constitution (deep, recursive).**
  - T0.1.1 `git -C constitution fetch --all --prune; pull; capture HEAD..@{u}`.
  - T0.1.2 Recurse: `git submodule foreach --recursive locate every nested constitution/governance copy; fetch/pull each`. Evidence: per-path log.
  - T0.1.3 Run `scripts/verify-all-constitution-rules.sh` (CONST-055) post-pull; record verdict.
- **P0.2 Audit project + submodules against ALL mandatory rules** (request §“Before we begin”).
  - T0.2.1 Run cascade verifier; enumerate any skipping/violation; open tracker items.
  - T0.2.2 Correct every detected violation (own sub-tasks per finding).
- **P0.3 Author the new fork-and-maintain anchor — §11.4.142** (next free number CONFIRMED by analysis-A; gate CM-COVENANT-114-142-PROPAGATION). Constitution submodule is **detached HEAD @ 60e2d66** → `git checkout main`

inside it FIRST; fix CLAUDE .md Revision-table drift (lags at Rev 23) per §11.4.44 in the same edit.

- T0.3.1 Use analysis-A's drafted anchor (forensic anchor = verbatim request lines 14–15; 6-clause body: fork+prefix → pointer-swap → recursive nested fork → automated upstream-merge → §11.4.28(B)-preserving edit allow-list [gitignore/remote-config/CI-disable/governance-pointers ONLY] → anti-bluff; composes §11.4.113 no-force-push).
- T0.3.2 Cascade into constitution/{Constitution,CLAUDE,AGENTS}.md + QWEN.md; consumer-side index updates.
- T0.3.3 Meta-test + paired §1.1 mutation for the propagation gate + 3 recommended per-family gates.
- **P0.4 Reusable generic scripts** in constitution/scripts/ (decoupled CONST-051; inherited-by-reference §11.4.80 — never copied). Names per analysis-A:
  - T0.4.1 fork\_third\_party\_submodule.sh (gh+glab fork-or-create under org+prefix caf-, params only, no force §11.4.113).
  - T0.4.2 update\_fork\_from\_upstream.sh (fetch upstream main/master → merge into fork → push fast-forward-only).
  - T0.4.3 resolve\_recursive\_fork\_deps.sh (recursive nested-dep handling; own-org → root per CONST-051; 3rd-party nested → fork).
  - T0.4.4 Anti-bluff Challenge + §1.1 mutation per script (throwaway repo bootstrap → assert remotes/merge).
- **P0.5 Commit + push constitution to ALL upstreams; bump pointer** (CONST-049 steps 4–7).

**Done when:** new anchor present + cascade-verified across fleet; scripts pass their Challenges with captured output; constitution pushed to all upstreams; meta-repo .gitmodules pointer bumped in same commit.

---

## SP1 — Model-access refinement

**Goal:** For every provider, an API key supplied OR auto-recognized from .env / .api\_keys / shell exports makes that provider's **working** models available — dynamically obtained, validated, verified via LLMsVerifier. Feeds: analysis ... - B - ....

**Verified facts (analysis-B):** verifier consumption is REAL (verifier/client.go live HTTP; adapter.GetVerifiedModels:183 cache→live→stale→fallback). But: secrets.LoadAPIKeys (loader.go:30, reads api\_keys.sh+.env) is **DEAD code** (zero production call-sites); no key-presence gate; the “working-model” predicate (Verified==true ∧ status=="verified" ∧ score≥GetMinAcceptableScore()) is loaded (adapter.go:175) but **never applied**; CLI handleListModels:1361 lists failed/pending as available. **The target flow already exists in helix\_agent** (verifier/provider\_types.go:349 SupportedProviders.EnvVars multi-alias + startup.go:389 per-key→DiscoverModels) + llms\_verifier/api\_keys/ → lift/extend per §11.4.74.

- **P1.1 Key auto-recognition.**
  - T1.1.1 **Wire** secrets.LoadAPIKeys **at startup** in cmd/server + cmd/cli/main.go BEFORE config load (§11.4.124 — wire, don't delete the dead code).

- T1.1.2 Multi-alias key-recognition table (reuse `helix_agent SupportedProviders.EnvVars`): `.env` → `.api_keys/api_keys.sh` → shell-exported precedence; per-provider detection.
  - T1.1.3 Tests (unit: precedence + absence; integration: real key → provider enabled).
  - **P1.2 Key-presence → provider-availability gate** — filter the verifier catalog by present keys (no key ⇒ provider hidden); fix `factory.go:9` (single-provider) → enumerate every key-present provider.
  - **P1.3 Dynamic working-model exposure via LLMsVerifier.**
    - T1.3.1 Apply the unused “working-model” filter (`key-present ∧ Verified==true ∧ status verified ∧ score≥min`); poll ≤60s (CONST-038); verified ≤24h (CONST-037).
    - T1.3.2 Replace hardcoded model lists (OpenAI:203, Anthropic:205, DeepSeek, Mistral, +~8) with verifier-sourced / dynamic fetch (CONST-036).
    - T1.3.3 Define safe verifier-disabled/cold behaviour — `FallbackModels` (CLI:1387) must NOT be shown as “working” without validation (else §11.4 PASS-bluff).
  - **P1.4 Docs/guides/manuals** for key setup + model availability + verification flow.
  - **P1.5 Test coverage** (unit/integration/e2e/Challenge) — real keys where available, anti-bluff liveness (§11.4.107).
- 

## SP2 — HelixAgent exposure extension

**Goal:** Under one root expose: AI debate ensemble + HelixLLM (if enabled) + **every** discovered provider individually + each provider’s working models; update provider SDKs to latest. Naming fits existing conventions. Feeds: analysis ...-C-....

**Verified facts (analysis-C):** exposure surface = `helix_agent/internal/router/router.go` (Gin /v1): ensemble POST /v1/ensemble/completions:676, providers GET /v1/providers:773 (reads `GetCapabilities().SupportedModels`), discovery :1306, verification :1378; runtime registry services.ProviderRegistry (provider\_registry.go:82). LLMProvider interface has **no** `GetModels()` — models only via `GetCapabilities().SupportedModels`. 48 hand-rolled HTTP providers, no vendor SDKs. HelixLLM buried as gated provider “helixllm” (`USE_HELIX_LLM=true`). “Working model” = `DiscoveredModel.Verified == true` (verifier). Building blocks exist; **no unified catalog joins ensemble + HelixLLM + each provider + each verified model**. Inner Go module is at `helix_code/go.mod` (not `helix_code/helix_code/go.mod`).

- **P2.0 Bluff remediation:** GET /v1/completion/models (handlers/completion.go:406) returns a **hardcoded 3-model list** → replace with verifier-sourced (CONST-036 / BLUFF-002). RED→GREEN + §11.4.135 guard.
- **P2.1 Unified catalog** joining ensemble + HelixLLM + each provider + each verified model as uniformly-named selectable targets (the missing layer).
- **P2.2 Per-provider individual exposure** under the root alongside ensemble + HelixLLM.
  - T2.2.1 Provider catalog entry per discovered provider (verifier-sourced).
  - T2.2.2 Per-provider working-model list (`Verified==true`); consider adding `GetModels()` to the provider interface.
  - T2.2.3 Naming scheme (analysis-C): `ensemble, ensemble/confidence_weighted, helixllm, helixllm/helixllm-default,`

- anthropic/claude-3-sonnet-20240229, openrouter/x-ai/grok-4 (reuses existing lowercase + vendor /model slash form).
  - T2.2.4 Promote HelixLLM to first-class root entry (not buried).
  - **P2.3 SDK/API currency** — deep web research (§11.4.99 latest-source) → update each provider SDK to latest (bedrockruntime/azcore/azidentity/tree-sitter pin; resolve gin 1.11.0[?]1.12.0 skew across helix\_code/helix\_agent); bridged CLI agents + their models up to date.
  - **P2.4 Docs/guides/manuals/graphs/schemes/SQL** + website + video-course updates (request-mandated deliverables).
  - **P2.5 Full test coverage.**
- 

## SP3 — Fork-ALL mechanism + auto-merge

**Goal:** Fork every cli\_agents/\* under vasic-digital/caf-\*, swap submodules to forks, recursively handle nested deps, auto-merge upstream regularly. Feeds: analysis ...-E-.... Uses SP0 scripts.

**Verified facts (analysis-E, parsed from 381-line .gitmodules):** 50 fork entries (49 top-level + 1 nested). **EXCLUSIONS:** cli\_agents/claude-code-source is already the operator's own GitLab mirror (milos85vasic/ccode-private) → **skip fork** (default-excluded). Only ONE nested submodule fleet-wide: cline/.gitmodules → cline-bench (HTTPS → SSH caf-cline-bench); max depth 1, no own-org nested chains (CONST-051 clean). OpenAI-Cookbook uses org-NNNN@github.com: alias (out of scope, resources dir) — parser must normalize.

- **P3.1 Fork mapping table** — built (49 GitHub upstreams → vasic-digital/caf-<name>, kebab names preserved); GitHub+GitLab dual-remote.
  - **P3.2 Execute fork-all** via the SP0 fork script (operator-gated; **high blast radius** → §11.4.101 **explicit confirm before any external mutation**; async fork-readiness polling).
  - **P3.3 Submodule swap** (.gitmodules URL rewrite + git submodule sync --recursive), per-fork validate buildable.
  - **P3.4 Nested-submodule resolution** (just cline-bench today; own-org → root per CONST-051; 3rd-party nested → fork).
  - **P3.5 Auto-merge automation** (scheduled merge-into-fork-only, **no-force** §11.4.113, clean commits §11.4.30).
  - **P3.6 Build + install each locally** with correct git-ignore hygiene (no stray files committed).
  - **P3.7 Reconcile script naming** between analysis-A (fork\_third\_party\_submodule.sh) and analysis-E (caf\_fork\_all.sh + caf\_lib.sh) into one unified, decoupled set before authoring.
- 

## SP4 — CLI-agent bridge providers (one provider per agent)

**Goal:** Each working CLI agent = its own provider proxying prompts via CLI + exposing power-features (Vision/Generative/RAG/Memory/MCP/LSP/ACP). Primary = system-installed; fallback = submodule-built. Re-export+install+validate every CLI config. Feeds: analysis ...-D-....

**Verified facts (analysis-D):** 8 agents system-installed NOW (claude/qwen/opencode/gemini/crush/codex/goose/copilot — versions captured). The real exec.LookPath orchestration lives in helix\_agent/internal/clis

(agents/base/base.go:120, instance\_manager.go), NOT in helix\_code/internal/llm (its copilot\_provider.go is HTTP/token, not CLI-exec). llm.Provider contract at missing\_types.go:356. Re-export exists (helixagent --generate-all-agents, main.go:4522).

- **P4.0 Pre-req remediation (anti-bluff, BLOCKS the rest).**
  - T4.0.1 **Fix stub per-agent packages** in helix\_agent/internal/clis that return hardcoded strings without exec (e.g. agents/qwencode/qwencode.go:101/115/137) — these are live §11.4 / CONST-035 bluffs. RED-on-stub → GREEN-on-real-exec (§11.4.115).
  - T4.0.2 Add replace dev.helix.agent wiring so HelixCode can import the clis layer (or thin adapter per CONST-051/§11.4.74).
- **P4.1 Per-agent provider contract** — one parametrized CLI-Agent-Provider implementing llm.Provider; reuse the clis layer via adapter, don't reimplement.
- **P4.2 Primary-vs-fallback selection** (real exec.LookPath system-installed → submodule-built).
- **P4.3 Power-feature passthrough** (RAG/Memory/MCP/LSP/ACP/Vision) + dynamic GetModels per agent (CONST-036) + capabilities from LLMsVerifier (CONST-040) — research per agent (§11.4.99): OpenCode/Zen non-interactive flags, Claude Code -p/--output-format, Qwen Code.
- **P4.4 Config re-export+install+validate** — re-export exists; **add the ABSENT filesystem-install + LIVE post-install validation** (each installed config drives a real proxied prompt → real result, captured evidence). Note: current configs point agents at HelixAgent (reverse) — D adds the forward bridge direction.
- **P4.5 Tier-1 agents first** (the 8 installed) → expand to ~43 candidates.
- **P4.6 Full test coverage** per agent (real proxied prompt → real result, anti-bluff §11.4.98 fully automated).

---

## SP5 — Parallel / subagent-driven enforcement

**Goal:** All HelixCode work can be done by coordinated parallel agents; subagents-driven used maximally without harming the main stream. Feeds: analysis ...-F-.... As a new HelixDevelopment submodule if warranted.

**Verified facts (analysis-F):** capability exists twice, duplicated not shared — HelixCode (internal/agent/coordinator.go:11-50 registry+queue+circuit-breakers+voting, agent/subagent/in-process+subprocess+worktree spawners, worker/SSH pool, workflow/DAG) vs HelixAgent (internal/agent/workflow.go:28 LangGraph-style, agents/swarm/, ensemble/, planning/{tree\_of\_thoughts,mcts,hiplan}). **GAP:** helix\_code/go.mod doesn't import the own-org agentic/concurrency/planning substrate; no universal “every operation is a parallel-dispatchable agent unit.”

- **P5.1 Map existing multi-agent/worker/queue capability.** (done — analysis-F)
  - **P5.2 De-duplicate into a shared substrate** (reuse before reimplement §11.4.74) — universal parallel-dispatchable agent-unit abstraction.
  - **P5.3 Coordination layer** (dispatch, isolation, quiescence §11.4.84, PWU §11.4.58, §11.4.103 ≥3 streams).
  - **P5.4 Wire into HelixCode + HelixAgent** (add the missing own-org imports).
  - **P5.5 Full test coverage** (stress/chaos §11.4.85 for concurrency).
-



## SP6 — Other execution models (dynamic flows, ...)

**Goal:** Dynamic flows + other execution models as NEW decoupled public submodules under HelixDevelopment, wired into HelixCode + HelixAgent. Feeds: analysis ... - F - ....

**Verified facts (analysis-F):** existing models = DAG workflow (HelixCode), graph/state-machine (helix\_agent/agent), ToT/MCTS/hierarchical planning.

**ABSENT as named packages:** pipeline, standalone dag, dataflow, reactive, saga, behavior-tree. Reuse candidates already in tree. Proposed NEW HelixDevelopment submodules: **flow\_engine**, **dag\_orchestrator**, **pipeline\_runtime**, **agent\_mesh/swarm\_kit**, **flow\_dsl**; 12 deep-research topics enumerated.

- **P6.1 Deep web research** on execution models (dynamic flow + others) — 12 topics in analysis-F (§11.4.99 latest-source).
  - **P6.2 Create public submodules** (gh+glab) under HelixDevelopment: flow\_engine / dag\_orchestrator / pipeline\_runtime / agent\_mesh / flow\_dsl; decoupled (CONST-051), helix-deps.yaml (CONST-054). Reuse existing own-org pieces first (§11.4.74).
  - **P6.3 Integration analysis + wiring** into both consumers (shared Node/Scheduler/Dispatcher/Resolver interfaces + go.mod replace).
  - **P6.4 Full test coverage + Challenges.**
- 

## SP7 — Testing & QA (cross-cutting)

**Goal:** 100% coverage across every supported test type + Challenges + HelixQA banks + fully-autonomous QA sessions; no bluff; independent-agent confirmation (CONST-050/048/098). Feeds: analysis ... - F - ....

**Verified facts (analysis-F):** HelixCode strong on unit/integration/e2e/security/chaos/stress/performance/benchmarking + §1.1 mutation meta-tests (tests/stresschaos/stresschaos\_meta\_test.go) + regression dir. **HelixCode-**

**local GAPS:** ddos, scaling, ux have no local harness (HelixQA/challenge-only); ui thin → delegation risks a coverage bluff unless evidence-backed. **⚠ HelixQA is STALE** (README round 219 / 2026-05-19).

- **P7.1 Update HelixQA** submodule to latest (fetch/pull + pointer bump).
  - **P7.2 Test-type coverage matrix** (unit/integration/e2e/full-automation/security/ddos/scaling/chaos/stress/performance/benchmark/ui/ux/Challenges/HelixQA) × every new feature.
  - **P7.3 Autonomous QA sessions** executing every registered bank with captured wire evidence.
  - **P7.4 Independent code-review agents** confirm each closure (§11.4.125/§11.4.134).
  - **P7.5 Regression-guard registration** for every fixed defect (§11.4.135).
- 

## 3. Cross-cutting mandatory constraints (apply to every SP)

- **Anti-bluff §11.4 family** — captured evidence on every closure; no metadata-only PASS.
- **No-force-push §11.4.113**; merge-onto-latest-main for every repo/submodule.
- **CONST-049** ordering for all constitution edits.
- **Subagent-driven default §11.4.70 / §11.4.103** — ≥3 parallel background streams while actionable items exist.

- **Fully-automated tests §11.4.98; rock-solid-proof-or-research §11.4.123.**
- **§11.4.122** — no silent removal of existing capabilities without operator confirmation.
- **CONST-046** — no hardcoded user-facing content; **CONST-036** — no hardcoded model lists.
- **Continuation §13.1 (CONST-044)** kept in sync each state-advancing commit.

## 4. Detailed implementation plans (written 2026-06-10)

| SP      | Plan doc                                          | Scope                                                                |
|---------|---------------------------------------------------|----------------------------------------------------------------------|
| SP1     | plans/2026-06-10-SP1-model-access-plan.md         | 6 phases / 18 TDD tasks / 4 decisions                                |
| SP2     | plans/2026-06-10-SP2-helixagent-exposure-plan.md  | 7 phases / 21 tasks / 6 decisions                                    |
| SP3     | plans/2026-06-10-SP3-fork-mechanism-plan.md       | 50-fork map + 3 scripts at scripts/caf / tasks T3.0–T3.7 / G-1 gated |
| SP4     | plans/2026-06-10-SP4-cli-bridge-plan.md           | 7 phases / 34 tasks / 4 decisions                                    |
| SP5+SP6 | plans/2026-06-10-SP5-SP6-parallel-dynamic-plan.md | reuse-first; create 2 repos / 5 decisions                            |
| SP7     | plans/2026-06-10-SP7-testing-qa-plan.md           | phases A–E / 6 gaps / 5 decisions                                    |

## 5. Consolidated decision register

**(a) Operator-GATED — irreversible / high-blast / genuinely ambiguous (I will NOT proceed without explicit go):** - **G-1** Execute **fork-ALL 49 repos** under `vasic-digital/caf - *` (SP3) — irreversible external. **✅ PRE-AUTHORIZED by operator 2026-06-10** — may run once the SP3 scripts pass their anti-bluff Challenge (no re-ask). - **G-2** Create **2 new HelixDevelopment repos** `dag_orchestrator + pipeline_runtime` (SP6) — irreversible external. **Still requires explicit go.** - **G-3 SDK-update blast radius** (SP2) — bumping `bedrockruntime/azcore/tree-sitter + resolving gin 1.11.0[?]1.12.0` across both modules can break builds → operator scope/sequence call. - **G-4 Video-course / website production scope** (SP2 OP-6) — defer vs in-scope. - **G-5** Whether to **de-bluff all ~70 agent packages now** vs tier-1-first + track-rest (SP4 D-C; D-6/D-9 live in `helix_agent`, a CONST-047 submodule).

**(b) Reversible / clear-default — I'll proceed on the subagents' recommended defaults unless you object (§11.4.101):** - Lift a decoupled key-recognition table from `helix_agent SupportedProviders.EnvVars` (SP1); `LoadAPIKeys` precedence = gap-fill (don't override already-exported shell env). - `Verifier-disabled` = honest empty + opt-in - - `include-unverified` live path (never show `FallbackModels` as “working”). - Unified catalog at `internal/catalog/ + GET /v1/catalog`; promote `HelixLLM` to first-class; naming `ensemble//helixllm//<provider>/<model>`. - De-harden funnel-first, then `OpenAI/Anthropic/DeepSeek/Mistral`. - SP5 substrate = `EXTEND Concurrency/Agentic, REUSE Planning, FOLD agent_mesh` in; first task adds the `replace` directives (closes D-7). **[M-1 fix] SP5 T5.2 is the SOLE owner of the replace dev.helix\_agent + replace dev.helix.substrate helix\_code/go.mod edits; SP4 T4.0.x CONSUMES them as a prerequisite (depends-on SP5 wiring) and NEVER re-edits go.mod** — resolves the parallel-

stream edit-collision. - New local harnesses for ddos/scaling/ux/ui with anti-delegation-bluff gates; HelixQA fetch-to-latest (§11.4.71 merge-onto-main).

## 5a. Bundle code-review (§11.4.142) — verdict + fixes applied

Independent review 2026-06-10-planning-bundle-review.md: **GO-WITH-FIXES** (0 BLOCKER · 1 MAJOR · 5 MINOR); 18 anchors spot-checked incl. all of D-1..D-9 — zero fabricated/drifted. Fixes: - **M-1 (MAJOR) — FIXED:** replace `dev.helix.agent/dev.helix.substrate` go.mod edits sole-owned by SP5 T5.2; SP4 consumes as prerequisite (see §5(b)). - **m-1 — N/A:** analysis-A's §11.4.142 choice is moot (SP0 constitution authoring descope; upstream already has §11.4.142). - **m-2 — noted:** SP4's "beyond roadmap" label for D-9 is cosmetic; D-9 is in the ledger (§1a). - **m-3 — TODO in SP7:** add a D-9 regression-guard row to the registry (carried into SP7 execution). - **m-4 — RESOLVED:** SP3 plan now written. - **m-5 — tracked:** CONST-055 full-rule audit = SP0.b (still unrun; offered to operator).

## 6. Next actions

1.  Planning complete: 6 analyses + 5 SP plans + defect ledger D-1..D-9. SP3 plan + planning-bundle code-review in flight.
2.  **HERE:** operator confirms the **decision register** (esp. the 5 gated items G-1..G-5).
3. Commit the planning bundle (needs §11.4.142 independent review GO + §11.4.75 .html/.pdf siblings + superproject pointer bump) — **operator: commit now?**
4. On go: implement RED-first per SP, each change through the §11.4.142 review gate; parallel non-contending streams (§11.4.103); irreversible externals only after G-1/G-2 go.